GCD of two numbers

R Vijayshri

July 2026

Calculating GCD of two unsigned integers

We use the Euclidean algorithm to calculate the gcd of two unsigned integers.

```
unsigned int gcd(unsigned int x, unsigned int y) {//we assume x>=y
    if (x==0) {
        return 0;
     }
    while (y != 0) {
        unsigned int temp = y;
        y = x % y;
        x = temp;
    }
    return x;
}
```

Correctness

We need to first show that gcd(x, y) = gd(y, x mod y), assuming $x \geq y$.

Theorem *Let $x \geq y$. Then, gcd(x, y) = gcd(y, x - y).*

Proof. Let d =gcd(x, y). Then $d|x$ and $d|y$.

$\implies d|(x - y)$

$\implies d|\text{gcd}(y, x - y)$.

But, the definition of d =gcd(a, b) is that $d|a$, $d|b$ and if there is some other number k such that $k|a$ and $k|b$, then $d|k$.

So, $gcd(y, x - y)|d$.

$\implies d$ =gcd$(y, x - y)$

□

Corollary *gcd(x, x mod y) = gcd$(x, x - y)$*

Proof. This follows from the fact that if $r = x \bmod y$, then $x = qy + r$, where $0 \leq r < y$. (Euclid's division algorithm).

So, applying the previous theorem q times, we get the result that gcd(x, x mod y) = gcd$(x, x - y)$. □

Time Complexity

Let x, y be both of n bits. We notice that $x \bmod y < \frac{x}{2}$. This is because if $y > \frac{x}{2}$, then $x \bmod y = x - y < \frac{x}{2}$. The result is trivially true if $> \frac{x}{2}$.

So, in every 2nd iteration of the while loop, the number of bits reduces by 1 for both x and y. So, in $2n$ iterations, y will be 0.

We also claim that $x \bmod y$ takes $O(n^2)$ time.

Since every 2nd iteration reduces the number of bits by 1, this algorithm takes $O(n^3)$ time.

References

- 1st Semester Programming and Data Structures Class notes.